Wave Equation

$n=1$, $n=3$ ✓

$n=2$: Method of Descent

Given $u(x_1, x_2, t)$,

let $\bar{x} = (x_1, x_2, x_3)$

$$\bar{u}(x_1, x_2, x_3, t) = u(x_1, x_2, t)$$

$$u_{tt} - u_{x_1x_1} - u_{x_2x_2} = 0$$

$$\bar{u}_{tt} - \bar{u}_{xx_1} - \bar{u}_{x_2x_2} - \bar{u}_{x_3x_3} = 0$$

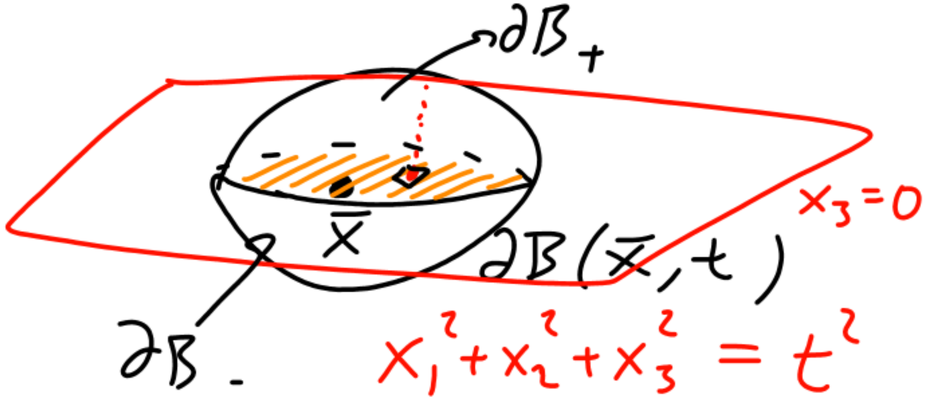
$$\bar{u}_{tt} - \Delta\bar{u} = 0$$

Kirchoff Formula:

$$\bar{u}(\bar{x}, t) = \frac{\partial}{\partial t}\left(t \oint_{\partial B(\bar{x},t)} \bar{g}\, dS\right) + t \oint_{\partial B(\bar{x},t)} \bar{h}\, dS$$

$$\boxed{\begin{aligned} u(x_1, x_2, 0) &= g(x_1, x_2) \\ u_t(x_1, x_2, 0) &= h(x_1, x_2) \\ \bar{g}(\bar{x}) &= g(x_1, x_2) \\ \bar{h}(\bar{x}) &= h(x_1, x_2) \end{aligned}}$$

Idea: write integral over $\partial B(\bar{x},t)$ in two pieces:



$$\gamma_+(x_1,x_2) = \sqrt{t^2-|x|^2}$$

$$\gamma_-(x_1,x_2) = -\sqrt{t^2-|x|^2}$$

$$x_1^2+x_2^2+x_3^2 = t^2$$

$$x_3 = \pm\sqrt{t^2-|x|^2} = \pm\gamma(x_1,x_2,t)$$

$$\int_{\partial B_+(\bar{x},t)} \bar{g}(\bar{x},t)\,dS = \int_{B(x,t)} g(x_1,x_2,t)\sqrt{1+|D\gamma_+|^2}\,dx_1\,dx_2$$

$$\frac{\partial \gamma_+}{\partial x_1} = \frac{1}{2\sqrt{t^2-|x|^2}}\cdot(-2x_1), \qquad |D\gamma_+|^2 = \frac{x_1^2+x_2^2}{t^2-|x|^2}$$

$$\sqrt{1+|D\gamma_+|^2} = \sqrt{1+\frac{x_1^2+x_2^2}{t^2-|x|^2}} = \sqrt{\frac{t^2}{t^2-|x|^2}}$$

$$\int_{\partial B_+(\bar{x},t)} \bar{g}(\bar{x})\,dS = \int_{B(x,t)} g(x) \frac{t}{\sqrt{t^2-|x|^2}}\,dx_1 dx_2$$

Southern hemisphere contributes the same term:

$$\int_{\partial B(\bar{x},t)} \bar{g}(\bar{x})\,dS = 2\int_{B(x,t)} g(x) \frac{t}{\sqrt{t^2-|x|^2}}\,dx \quad \Bigg| \quad \fint_{\partial B(\bar{x},t)} \bar{g}(\bar{x})\,dS = \frac{1}{2\pi t}\int \frac{g(x)\,dx}{\sqrt{t^2-|x|^2}}$$

Overall:

$$\bar{u}(\bar{x},t) = \frac{\partial}{\partial t}\left(\frac{1}{\pi}\int_{B(x,t)} g(x) \frac{1}{\sqrt{t^2-|x|^2}}\,dx\right) + \int_{B(x,t)} \frac{h(x)\,dx}{\sqrt{t^2-|x|^2}}$$

Since $u(x,t) = \bar{u}(\bar{x},t)$, this is a formula for u.

Compare the solution formulas for 2D & 3D Wave Equations:

3D: $$u(x,t) = \fint_{\partial B(x,t)} \left[t h(y) + g(y) + Dg(y)\cdot(y-x) \right] dS$$

2D: $$u(x,t) = \frac{\partial}{\partial t}\left(\frac{1}{\pi} \int_{B(x,t)} \frac{g(x)\, dx}{\sqrt{t^2 - |x|^2}} \right) + \int_{B(x,t)} \frac{h(x)\, dx}{\sqrt{t^2 - |x|^2}}$$

Huygen's Principle: 3D Wave Propagation is over boundary of $B(x,t)$

2D Wave Propagation is over interior.

Nonhomogeneous Wave Equation:

$$\begin{cases} u_{tt} - \Delta u = f(x,t) \\ u(x,0) = 0 \\ u_t(x,0) = 0 \end{cases}$$

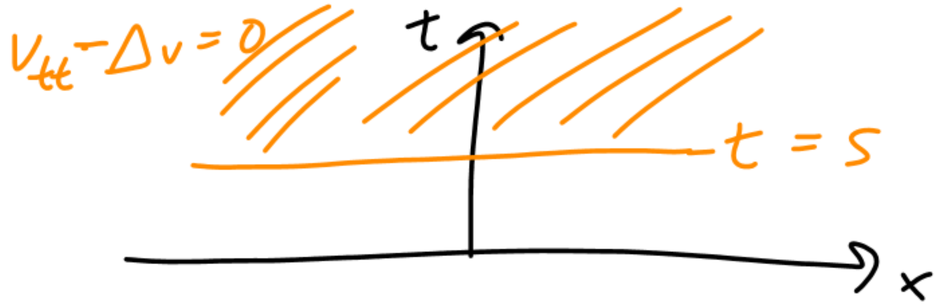
Duhamel's Principle:

Let $v(x,t,s)$ solve

$$v_{tt} - \Delta v = 0 \quad \text{in } \mathbb{R}^n_x \times \{t \geq s\}$$

$$v(x,t,t) = 0$$

$$v_t(x,t,t) = f$$



Let

$$u(x,t) = \int_0^t v(x,t,s)\,ds.$$

Check:

$$u_t = \overset{=0}{v(x,t,t)} + \int_0^t v_t(x,t,s)\,ds$$

$$u_{tt} = v_t(x,t,t) + \int_0^t v_{tt}(x,t,s)\,ds$$

$$= f(x,t) + \int_0^t v_{tt}(x,t,s)\,ds$$

$$\Delta u(x,t) = \int_0^t \Delta v(x,t,s)\,ds$$

$$u_{tt} - \Delta u = f(x,t) + \int_0^t (v_{tt} - \Delta v)\,ds = f(x,t)$$

$n=1$

$v(x,t)$ solves

$$v_{tt} - v_{xx} = 0 \text{ in } (t \geq s) \times \mathbb{R}_x$$

$$v(x,s) = 0$$

$$v_t(x,s) = f(x,s)$$

d'Alembert:

$$v(x,t,s) = \frac{1}{2}\int_{x-(t-s)}^{x+t-s} f(y,s)\,dy$$

$$u(x,t) = \int_0^t v(x,t,s)\,ds$$

$$= \frac{1}{2}\int_0^t \int_{x-t+s}^{x+t-s} f(y,s)\,dy\,ds$$

$\tau = t - s \quad ds = -d\tau$

$$\hookrightarrow = \frac{1}{2}\int_0^t \int_{x-\tau}^{x+\tau} f(y,t-\tau)\,dy\,d\tau \quad \checkmark$$

$n = 3$:

$$v(x,t,s) = \fint_{\partial B(x,t-s)} (t-s) f(y,s)\, dS$$

$h(y)$

$g(y) = 0$

$$u(x,t) = \int_0^t \fint_{\partial B(x,t-s)} (t-s) f(y,s)\, dS\, ds$$

$$= \int_0^t \frac{1}{4\pi(t-s)} \int_{\partial B(x,t-s)} f(y,s)\, dS\, ds$$

$\tau = t - s$

$d\tau = -ds$

$$= \int_0^t \frac{1}{4\pi\tau} \int_{\partial B(x,\tau)} f(y, t-\tau)\, dS\, d\tau$$

In general case,

$$\begin{cases} u_{tt} - u_{xx} = f \\ u(x,0) = g \\ u_t(x,0) = h \end{cases},$$

we can add together homogeneous & non-homogeneous solution formulas.

Energy Methods

Thm (Uniqueness on Bounded Domain)

If $U \subseteq \mathbb{R}^n$ is a bounded domain, there exists at most one smooth solution to

$$\begin{cases} u_{tt} - \Delta u = f(x,t) & x \in U,\ t > 0 \\ u(x,0) = g(x) \\ u_t(x,0) = h(x) \\ u(x,t) = 0, & x \in \partial U. \end{cases}$$

Pf Define

$$E[u] = \int_U \left[u_t^2 + |Du|^2 \right] dx$$ for any solution u.

Given two solutions u, v,

let $w = u - v$

Then

$$\begin{cases} w_{tt} - \Delta w = 0 \\ w(x,0) = 0 \\ w_t(x,0) = 0 \\ w(x,t) = 0, \quad x \in \partial U \end{cases}$$

Claim: $E[w] = \int_U \left[w_t^2 + |Dw|^2 \right] dx$ is constant in t.

$$\frac{d}{dt} E[w] = \int_U \left[2 w_t w_{tt} + 2 Dw \cdot Dw_t \right] dx$$

$$\int_U Dw \cdot Dw_t \, dx = -\int_U w_t \Delta w \, dx + \cancel{\int_{\partial U} w_t Dw \cdot \nu \, dS}^{\,0}$$

$w_t = 0$ on ∂U

$$\frac{d}{dt} E[w] = 2 \int_U w_t (w_{tt} - \Delta w) \, dx = 0.$$

At $t=0$,

$$E[w(x,0)] = \int_U \left[w_t^2 + |Dw|^2\right] dx$$

$$= 0, \quad w_t = 0, \quad Dw = 0$$

$\Rightarrow E(w,t) = 0$ for all t.

$\Rightarrow w_t = 0$, $Dw = 0$ for all x, t

$\Rightarrow w = \text{constant} \Rightarrow w = 0$ everywhere

$\Rightarrow u = v$.

(Could have argued:

$$w_{tt} - \Delta w = 0$$

$$\int w_t (w_{tt} - \Delta w)\, dx = 0$$

$$\frac{1}{2}\int \left[\frac{d}{dt}(w_t^2) + 2Dw_t \cdot Dw\right] dx = 0$$

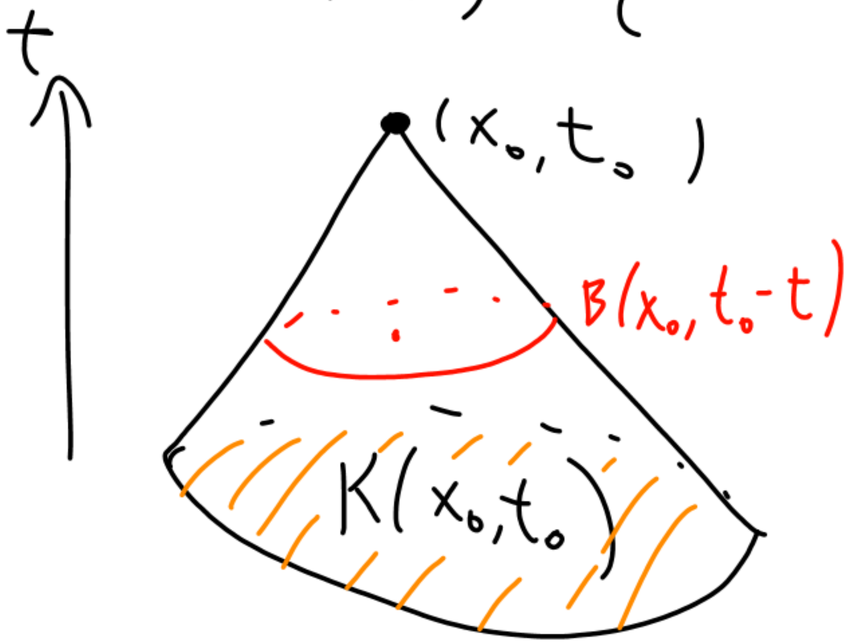
$$\frac{1}{2}\frac{d}{dt}\int \left(w_t^2 + |Dw|^2\right) dx = 0$$

Alternate Proof of Finite Speed of Propagation:

Define

$$K(x_0,t_0) = \{(x,t) : 0 \le t \le t_0,\ |x-x_0| \le (t-t_0)\}$$

Backward Wave Cone.



<u>Thm</u> If $u_{tt} - \Delta u = 0$ and $u(x,0) = 0$ in $B(x_0,t_0)$, then

$$u = 0 \text{ in } K(x_0,t_0)$$

<u>Pf</u> Define

$$e(t) = \int_{B(x_0, t_0 - t)} \left[u_t^2 + |Du|^2 \right] dx$$

$e(t)$ is a "local energy".

$$e'(t) = -\int_{\partial B(x_0, t_0 - t)} \left[u_t^2 + |Du|^2 \right] dx + \int_{B(x_0, t_0 - t)} \left[2 u_t u_{tt} + 2 Du \cdot Du_t \right] dx$$

$$= I_1 + I_2 .$$

$$I_2 = 2\int_{B(x_0,t_0-t)} \left[u_t u_{tt} + Du \cdot Du_t\right] dx$$

$$= \cancel{2\int_{B(x_0,t_0-t)} \left[u_t u_{tt} - \Delta u \, u_t\right] dx} + 2\int_{\partial B(x_0,t_0-t)} u_t \, Du \cdot \nu \, dS$$

$$\overset{\shortparallel}{u_t\left[u_{tt} - \Delta u\right] = 0}$$

$$e'(t) = I_1 + I_2$$

$$= \int_{\partial B(x_0,t_0-t)} -\left[u_t^2 + |Du|^2\right] + \left[2 u_t \, Du \cdot \nu\right] dS$$

$$|u_t \, Du \cdot \nu| \le |u_t| \cdot |Du|$$
$$\le \frac{1}{2} u_t^2 + |Du|^2$$

$ab \le \frac{1}{2}(a^2+b^2)$

$$\rightarrow e'(t) \le \int_{\partial B(x_0, t_0 - t)} \left[-u_t^2 - |Du|^2 + u_t^2 + |Du|^2 \right] dx$$

$$\rightarrow e'(t) \le 0$$

Since $u = 0$ in $B(x_0, t_0)$, $e(0) = 0$

$e(t) \ge 0$ for all t $\rightarrow$ $e(t) = 0$ for $t \in [0, t_0]$

$\rightarrow u = 0$ in $K(x_0, t_0)$.